6G5Z3011 MULTI-VARIABLE CALCULUS AND ANALYTICAL METHODS

TUTORIAL SHEET 05

Qs 1 – 5 on **integrals over rectangular regions**, Qs 6 - 8 on **integrals over non-rectangular regions and transforming coordinates**

(1) Evaluate the following double integral

$$\int_1^2 \int_0^3 x^2y + y^2x \ \ dx\, dy.$$

What does this integral represent?

(2) Sketch the region over which the double integral below is taken and hence evaluate it.

$$\int_{-1}^1 \int_{-2}^2 4xy + \sin x + \cos y \ \ dx\, dy.$$

(3) Sketch the regions of integration for the following two integrals and hence rewrite the sum as a single integral and evaluate it.

$$\int_0^3 \int_1^3 9x^2y^2 + 4xy + 5 \ \ dx\, dy + \int_0^1 \int_0^3 9x^2y^2 + 4xy + 5 \ \ dy\, dx.$$

(4) Evaluate the following double integrals by first reversing the order of integration.

(a)

$$\int_0^1 \int_0^{0.5} ye^{xy} \ \ dy\, dx$$

(b)

$$\int_0^1 \int_0^{0.5} x \sin(xy) \ \ dx\, dy$$

(5) Find the average height of the surface defined by $z = x^2 + y^2$ that lies above the square bounded by the lines $x = 1$, $x = -1$, $y = 1$ and $y = -1$.

(6) Sketch the region over which the following double integral is taken and hence evaluate it.

$$\int_1^2 \int_{-x}^x \frac{y}{x} + 1 \ \ dy\, dx$$

(7) Evaluate the following integrals by first reversing the order of integration.

(a)

$$\int_0^1 \int_0^{\cos^{-1} y} \sec x \ \ dx\, dy$$

(b)

$$\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} \ \ dx\, dy,$$

(Hint: make use of the substitution $u = x^2$ towards the end)

(8) Evaluate the following double integrals by first transforming to polar coordinates.

(a)

$$\int_0^2 \int_0^{\left(4-y^2\right)^{\frac{1}{2}}} \left(x^2 + y^2\right)^{\frac{5}{2}} \tan^{-1}\left(\frac{y}{x}\right) \ \ dx\, dy$$

(b)

$$\int_0^\infty \int_0^\infty \frac{1}{\left(x^2 + y^2 + 1\right)^2} \ \ dx\, dy$$